

TEL-TSI

Introduction to semantic modelling

A **concept scheme** (in SKOS: *ConceptScheme*) is a structured collection of concepts that carry a fixed, shared meaning within a particular domain. The goal is for everyone working with the model — people and systems alike — to use the same words for the same things. That sounds obvious, but in practice language causes a surprising amount of confusion: what one organisation calls a *measurement point*, another calls a *sensor*, a *location*, or an *observation station*.

A concept scheme solves this by recording, for each concept:

- a **preferred label** (prefLabel) it is known by,
- a **definition** that unambiguously describes what the concept does and does not cover,
- and optionally a **scope note** or **example** to clarify borderline cases.

Concepts rarely stand on their own. They are related **hierarchically** (a vehicle is broader than a truck) or **associatively** (a road user relates to a road segment). These relationships make it possible for queries and analyses to navigate between general and specific concepts, and to make connections between concepts explicit. Within SKOS we use `broader`, `narrower`, and `related` for that.

When a concept also exists elsewhere — in a national or European standard, for instance — we record that as a **mapping relation**. This prevents reinventing the wheel: we link to the source so interoperability with external systems remains possible.

This document describes the concept scheme as it is currently recorded. The following sections list the metadata, every concept, the relationships between them, and — where the model is connected to a data source — how it is linked to that source data.

TEL-TSI

TEL TSI

Metadata

Field	Value
URI	https://github.com/NAPCORE/its-data-dictionary/tel-tsi
Version	1.0.0
Publication status	Draft
Management status	Active
Organization	NAPCORE
Created on	2026-05-21
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Last modified by	Arthur Keesen

Model overview

Visual overview of the concepts and their relations. Hierarchical relations are drawn with solid arrows; associative relations are drawn with dashed arrows.



Concepts

Concepts overview

Concept	Definition	Source
Reference train identifier	Object that uniquely identifies a planned train transport (PlannedTransportIdentifier, type TR), specified by the lead railway undertaking, used as the reference for a train across networks.	
Train object	Object (TrainID) identifying a train, specified by the lead railway undertaking, to which operational train running data is related.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_13
Path request object	Object (PathRequestID) identifying a request for infrastructure capacity, specified by the lead railway undertaking; carries the operational train number and commercial train number as attributes.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_13
Path object	Object (PlannedTransportIdentifier, type PA) identifying an allocated train path, specified by the infrastructure manager.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_13
Route object	Object (PlannedTransportIdentifier, type RO) identifying a route, specified by the lead railway undertaking.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_13
Case reference object	Object (PlannedTransportIdentifier, type CR) identifying a case reference, specified by the holder of the data.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_13
PathRequestMessage	Message by which an applicant requests infrastructure capacity (a train path) from the infrastructure manager, in accordance with the ERA Ontology.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_10
PathDetailsMessage	Message by which the infrastructure manager provides the details of a path in response to a path request, in accordance with the ERA Ontology.	

Concept	Definition	Source
		http://data.europa.eu/eli/reg_impl/2026/253/oj#art_10
PathConfirmedMessage	Message by which the applicant confirms the path details proposed by the infrastructure manager.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_10
PathDetailsRefusedMessage	Message by which the applicant refuses the path details proposed by the infrastructure manager.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_10
PathCanceledMessage	Message by which a previously confirmed or requested path is cancelled.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_10
PathNotAvailableMessage	Message by which the infrastructure manager notifies that a requested path is not available.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_10
ReceiptConfirmationMessage	Message confirming receipt of a message between telematics stakeholders.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_10
TrainCompositionMessage	Message conveying the composition of a train, including vehicle-level information, total train mass, European Vehicle Number of traction units, and flags for dangerous goods, intermodal loading units, armed forces and quieter routes.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_11
PassengerTrainCompositionMessage	Message conveying the composition of a passenger train, including seating maps, classes, coaches for persons with reduced mobility and bicycle spaces.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_11
TrainReadyMessage	Message by which the railway undertaking notifies that a train is ready for departure; status values include 'NotReady' with sub-fields for train delay, train ready time and delay cause.	

Concept	Definition	Source
		http://data.europa.eu/eli/reg_impl/2026/253/oj#art_11
TrainRunningInformationMessage	Message conveying actual train running information at reporting points.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_11
TrainRunningForecastMessage	Message conveying forecast train running information, with expected accuracy as set out in the regulation's accuracy methodology.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_11
TrainDelayCauseMessage	Message conveying the cause of a train delay.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_11
TrainRunningInterruptionMessage	Message notifying an interruption to the running of a train.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_11
Working timetable	The working timetable data as defined in Article 3(28) of Directive 2012/34/EU, published as open data under a Creative Commons BY-ND 4.0 licence.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_3.pnt_29
Historic record of train data	Retained record (at least 12 months) of train data, containing the working timetable, reference train identifier, commercial train number, reporting locations, actual times, delay and delay cause, and train composition.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_11
ConsignmentNoteMessage	Message conveying the electronic consignment note (eCN), i.e. the transport document referred to in Article 6 of Council Regulation No 11/1960, sealed with a qualified electronic seal pursuant to Regulation (EU) No 910/2014.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_14
WagonRunningInformationMessage	Message conveying running information about a freight wagon across its movement life cycle; decomposed into the individual wagon events specified in the Annex.	

Concept	Definition	Source
		http://data.europa.eu/eli/reg_impl/2026/253/oj#art_14
Intermodal loading unit movement	Data conveying the movement of an intermodal loading unit.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_14
Wagon forecast information	Forecast information for a freight wagon, with accuracy determined by the regulation's accuracy methodology.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_14
Geo-localisation of freight wagons	Wagon event conveying the geographic position of a freight wagon.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_14
Wagon status	Wagon event conveying the current status of a freight wagon.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_14
Wagon ready for movement	Wagon event indicating that a freight wagon is ready to be moved.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_14
Wagon pulled	Wagon event indicating that a freight wagon has been pulled.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_14
Wagon departure from origin	Wagon event indicating departure of a freight wagon from its origin.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_14
Wagon arrival at intermediate location	Wagon event indicating arrival of a freight wagon at an intermediate location.	

Concept	Definition	Source
		http://data.europa.eu/eli/reg_impl/2026/253/oj#art_14
Wagon departure from intermediate location	Wagon event indicating departure of a freight wagon from an intermediate location.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_14
Wagon handed over	Wagon event indicating that a freight wagon has been handed over between responsible parties.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_14
Wagon taken over	Wagon event indicating that a freight wagon has been taken over by a responsible party.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_14
Wagon irregularity	Wagon event reporting an irregularity affecting a freight wagon.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_14
Wagon irregularity rectified	Wagon event indicating that a previously reported wagon irregularity has been rectified.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_14
Wagon arrival at destination	Wagon event indicating arrival of a freight wagon at its destination.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_14
Wagon delivered	Wagon event indicating that a freight wagon has been delivered.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_14
Rolling stock reference databases	Reference databases for rolling stock, including the European Vehicle Number, keeper, type and technical parameters; managed by the rail sector and linked from the ERA common central repository.	

Concept	Definition	Source
		http://data.europa.eu/eli/reg_impl/2026/253/oj#art_8
Intermodal loading unit reference databases	Reference databases for intermodal loading units, including ILU code, type, dimensions and owner.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_8
Wagon and intermodal loading unit operational databases	Operational databases recording the operational states of freight wagons and intermodal loading units across their movement.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_8
Loading of the freight wagon	Operational state indicating that a freight wagon is being loaded.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_8
Loaded freight wagon on journey	Operational state indicating that a loaded freight wagon is on its journey.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_8
Empty freight wagon on journey	Operational state indicating that an empty freight wagon is on its journey.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_8
Unloading of the freight wagon	Operational state indicating that a freight wagon is being unloaded.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_8
Empty freight wagon under fleet management control	Operational state indicating that an empty freight wagon is under fleet management control.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_8
Dossier reference	Reference identifying a ticketing dossier, used to link tickets, reservations and after-sale processes.	

Concept	Definition	Source
		http://data.europa.eu/eli/reg_impl/2026/253/oj#art_3.pnt_3
Availability and reservation request message	Message by which availability is checked and a reservation is requested for a rail product.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_17
Availability and reservation response message	Message conveying the response to an availability and reservation request.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_17
PRM-assistance availability and reservation	Message for checking availability of and reserving assistance for persons with disabilities or reduced mobility.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_17
Bicycle-place availability and reservation	Message for checking availability of and reserving a bicycle place.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_17
Car-carriage availability and reservation	Message for checking availability of and reserving the carriage of cars, motorcycles or boats.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_17
Ticket annotation	After-sale message recording an annotation on a ticket.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_17
Reimbursement or compensation request	After-sale message requesting reimbursement or compensation in respect of a ticket.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_17
Passenger timetable		

Concept	Definition	Source
	The date and time of passenger rail transport services operated as a commercial service by a railway undertaking during a given time interval, used in the context of rail ticketing and rail passenger travel information referred to in Section 4 of the Annex.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_3.pnt_33
Minimum connection time	The minimum amount of time that is sufficient for a passenger to connect between the access nodes of two consecutive passenger transport services included in a journey.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_3.pnt_24
Tariff	Aggregated data established by a railway undertaking (or data holder) providing all relevant tariff details, including the rail-transport-service price due by a distributor to the railway undertaking and the associated attributes and conditions.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_3.pnt_43
Offer	Aggregated data established by a distributor combining applicable tariffs, with the rail-product price due by the retailer to the distributor.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_3.pnt_44
Yielded price	Price associated with a tariff resulting from an algorithm based on variables, referred to as the yielding rule.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_3.pnt_45
Rules for ticket issuing and ticket inspections	Data establishing the rules governing the issuing of tickets and the inspection of tickets.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_17
Information on online sales channels	Data describing the online sales channels through which rail products are offered.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_17
Conditions of carriage	Data establishing the conditions of carriage, decomposed for passengers, persons with reduced mobility (with rolling-stock and station accessibility), luggage, bicycles, and cars/motorcycles/boats.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_17
Types and formats of tickets	The catalogue of ticket types and formats, including secured-paper, blank-paper and mobile/flexible-size ticket layouts with their associated security models.	

Concept	Definition	Source
		http://data.europa.eu/eli/reg_impl/2026/253/oj#art_17
Common reference data for rail ticketing	Code lists for ticketing managed and assigned by the Agency, used as a basis for data sharing in the rail ticketing domain.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_9
Security elements for electronic delivery	Security elements used for the electronic delivery of tickets.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_17
Train departures information	Data on train departures for display at stations.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_18
Train arrivals information	Data on train arrivals for display at stations.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_18
Deviations information	Data on deviations from the scheduled service for display at stations.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_18
Information in the vehicle	Passenger travel information provided inside the vehicle.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_18
Distributor	Entity with the technical capacity to combine rail products from one or more railway undertakings to supply an issuer or retailer, with rights to operate without an initial query to attributing systems and autonomous rights to check availability of price.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_3.pnt_38
Issuer	Entity with the legal and technical capacity to issue, for a retailer, a ticket on which the parties to the contract of carriage are indicated.	

Concept	Definition	Source
		http://data.europa.eu/eli/reg_impl/2026/253/oj#art_3.pnt_40
Retailer	Entity that combines and/or sells rail products from one or more distributors, obtaining access via an initial request; it may apply fees or reductions but cannot amend rail-product content.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_3.pnt_42
Attributing system	The railway undertaking's digital system hosting the online catalogue of individual rail transport services and enabling distributors to check availability and make reservations.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_3.pnt_41
Journey planner	A search engine used to find an optimal journey based on defined criteria set out by its user.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_3.pnt_39
Ticket control organisation	An organisation empowered to inspect passenger tickets, including at platform access doors.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_3.pnt_46
Rail product	A tailored service in relation to a passenger rail transport service linked to a tariff.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_3.pnt_4
Meta station	A group of stations within a functional urban area, including one or more urban nodes listed in Annex II to Regulation (EU) 2024/1679, that may extend across multiple Member States and where distinct stations can be reached by connecting urban transport modes and urban transport services, whether scheduled or unscheduled.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_3.pnt_30
Data	A machine-readable representation of acts, facts or information and any compilation of such acts, facts or information intended to be shared in a harmonised and structured format.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_3.pnt_1
Data sharing		

Concept	Definition	Source
	The provision of data by a data holder to a data recipient for the purpose of the joint or individual use of such data, directly or through an intermediary, for example under open or commercial licences subject to a fee or free of charge.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_3.pnt_2
Rail ticketing	The process comprising at least one of the following for passenger rail transport services: the digital presentation of services for journey planning; the digital checking of availability or reservation of rail products for sales; the digital issuing and control of tickets; the digital handling of after-sale processes.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_3.pnt_3
Rail transport service	A scheduled service for carriage by rail between a point of departure and a point of arrival, covering transport services for goods, passengers, luggage, bicycles or cars, and tailored transport arrangements, as well as passenger assistance including services for re-routing and journey continuation.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_3.pnt_7
Access to data	The right to consult the data with or without control over it, distinctively from other rights and obligations relating to its use for non-commercial or commercial purposes.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_3.pnt_14
Multi-network process	A process that is affected by the operation of a rail transport service across two or more networks managed by different infrastructure managers or operators of rail freight service facilities.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_3.pnt_20
Direct rail transport service	A rail transport service that may include intermediate commercial stops between its origin and destination but does not involve any transfer of passengers or freight between trains.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_3.pnt_21
Application programming interface	A set of functions, procedures, definitions and protocols for interoperable data sharing between telematics applications.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_3.pnt_22
Journey	An end-to-end trip of a passenger or a load that can include multiple legs.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_3.pnt_26
Public departure time	The boarding time limit for passengers, that is the point in time when access to the train, platform or station is closed in preparation for departure.	

Concept	Definition	Source
		http://data.europa.eu/eli/reg_impl/2026/253/oj#art_3.pnt_27
Public arrival time	The alighting time for passengers, that is the point in time when the doors of the train are opened at the destination platform or terminal.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_3.pnt_28
Data holder	A natural or legal person that has the right to use or an obligation to produce and make available data.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_3.pnt_12
Data recipient	A natural or legal person that has the right to access the data produced and made available by the data holder under specified access conditions for use.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_3.pnt_13
Intermodal transport operator	An entity that organises freight transport services connecting two or more freight terminals with more than one transport mode.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_3.pnt_36
Freight customer	The entity that has issued consignment information to the freight railway undertaking and has concluded a contract of carriage; may be a logistics service provider, intermodal transport operator, freight forwarder, fleet manager of empty wagons, or the data holder.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_3.pnt_35
ERA Ontology	A technical document issued by the Agency pursuant to Article 4(8) of Directive (EU) 2016/797, setting out human-readable and machine-readable data definitions and presentations and the associated quality and accuracy requirements for data elements of the Union rail system.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_3.pnt_16
Data element	A field, value, or attribute of the data shared, structured consistently with the overall protocol used for data sharing and carrying essential information that contributes to the meaning and purpose of the data shared, enabling data sharing between systems, applications or organisations.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_3.pnt_17
Reference data	A data set of standardised and interoperable single identifiers to be used as a basis for data sharing in a specific domain.	

Concept	Definition	Source
		http://data.europa.eu/eli/reg_impl/2026/253/oj#art_3.pnt_18
Metadata	Metadata within the meaning of Article 2(2) of Regulation (EU) 2023/2854 (Data Act).	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_3.pnt_19
Data quality	The degree of fitness for use at which the characteristics of data satisfy the stated and implied level of accuracy, completeness, consistency, timeliness and uniqueness of datasets, and create actionable insights for other users under specified conditions.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_3.pnt_47
Accuracy (data quality dimension)	Data quality dimension expressing the degree to which data correctly represents the real-world value it is intended to represent.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_8
Completeness (data quality dimension)	Data quality dimension expressing the degree to which all required data is present.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_8
Consistency (data quality dimension)	Data quality dimension expressing the degree to which data is free of contradiction across datasets and over time.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_8
Timeliness (data quality dimension)	Data quality dimension expressing the degree to which data is available within the required time.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_8
Uniqueness (data quality dimension)	Data quality dimension expressing the degree to which there is no unwanted duplication of data.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_8
Organisation code	Single identifier of any organisation or role involved in rail data sharing, assigned by the Agency.	

Concept	Definition	Source
		http://data.europa.eu/eli/reg_impl/2026/253/oj#art_9
Location code	Single identifier of every geographical point relevant to rail data sharing, assigned by the Agency.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_9
Virtual location code	Code assigned to non-rail-network locations used by rail-replacement transport services, services sold as rail but not operated on rail, and meta stations.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_9
EU-Id	The canonical European identifier used by data holders when feeding the national access points.	http://data.europa.eu/eli/reg_impl/2026/253/oj#art_9